



FinSight Bank — NPA Prediction & Credit Risk Modelling

Executive Summary

The objective of this project is to analyse loan data and identify factors that contribute to loan defaults (Non-Performing Assets - NPA). The project helps FinSight Bank improve credit risk management by identifying high-risk customers and supporting better lending decisions.

The analysis was performed using loan data collected from multiple sources. Various data cleaning, exploratory data analysis (EDA), statistical analysis, feature engineering, and regression modelling techniques were applied to derive meaningful insights and business recommendations.

1. Data Collection and Integration

The dataset consisted of loan information stored across 9 different CSV files. These files contained customer details, loan information, repayment records, credit bureau data, and other related information.

All datasets were merged using the common key **loan_id** to create a single analytical dataset.

Key Activities

- Loaded data from 9 CSV files.
 - Merged all tables using loan_id.
 - Verified row counts after merging.
 - Checked for missing or unmatched records.
 - Optimized memory usage by converting data types where required.
-

2. Data Cleaning and Preprocessing

Data cleaning was performed to improve data quality and ensure accurate analysis.

Handling Missing Values

Missing values were identified and treated using appropriate techniques.

Methods Used

- Median imputation for numerical variables.
- Domain-based imputation for specific business attributes.
- Special category values assigned where required.

Handling Invalid Data

Several data quality issues were identified and corrected.

Examples:

- Negative interest rates.
- Zero annual income values.
- Invalid debt-to-income ratios.

- Incorrect or inconsistent entries.

Duplicate Check

Duplicate records were identified and removed to maintain data integrity.

Outlier Treatment

Extreme values were treated using Winsorization to reduce their influence without removing valid observations.

3. Exploratory Data Analysis (EDA)

EDA was performed to understand the data distribution, identify patterns, and discover important risk factors.

Default Rate Analysis

The overall loan default rate was found to be 3.88%.

Interpretation

Most borrowers successfully repay their loans, while a small percentage become NPAs.

CIBIL Score Analysis

The average CIBIL score of defaulted customers was lower than that of non-defaulted customers.

Interpretation

Customers with lower CIBIL scores have a higher probability of default.

Loan Grade Analysis

Credit grades showed a strong relationship with default rates.

Findings

Grade	Risk Level
A	Very Low
B	Low
C	Moderate
D	High
E	Very High
F	Extremely High
G	Highest

Interpretation

As credit grade decreases, the probability of loan default increases.

Interest Rate Analysis

Defaulted borrowers generally had higher interest rates compared to performing borrowers.

Interpretation

Higher interest rates often indicate higher borrower risk.

State-wise Default Analysis

Default rates varied across states.

Interpretation

Geographical factors may influence repayment behavior and credit risk.

COVID-19 Impact Analysis

The default rate increased during 2020 compared to previous years.

Interpretation

Economic disruptions caused by the COVID-19 pandemic negatively affected loan repayment performance.

4. Statistical Analysis

Statistical methods were used to validate findings and understand relationships between variables.

Correlation Analysis

A correlation heatmap was used to identify relationships among numerical variables.

Purpose

- Identify highly related variables.
 - Detect multicollinearity.
 - Support feature selection.
-

Hypothesis Testing

A t-test was conducted to compare LGD percentages before and during the COVID-19 period.

Result

No statistically significant difference was found between the two groups.

Interpretation

COVID-19 increased loan defaults, but it did not significantly affect loss severity in this dataset.

5. Feature Engineering

New features were created to improve predictive performance.

Features Created

Repayment Burden Features

- EMI to Income Ratio
- Loan to Income Ratio

Credit Behaviour Features

- Credit Utilization Score
- Delinquency Severity Score

Financial Stability Features

- Income Stability Ratio
- Credit Depth Score

Log Transformations

Log transformations were applied to highly skewed variables such as:

- Annual Income
- Loan Amount

Purpose

- Reduce skewness.
- Improve regression performance.
- Create a more normal distribution.

6. Regression Modelling

Regression techniques were applied to predict Loss Given Default (LGD).

Models Used

1. Linear Regression
2. Ridge Regression
3. Lasso Regression
4. Elastic Net Regression

Evaluation Metrics

R-Squared (R^2)

Measures how much variation in the target variable is explained by the model.

RMSE (Root Mean Square Error)

Measures prediction error.

Findings

The models achieved very low R^2 scores, indicating that the current set of features had limited ability to predict LGD accurately.

This suggests that additional variables and advanced modelling techniques may be required.

7. Key Findings

1. Lower CIBIL scores are associated with higher default risk.
 2. Higher interest rates are linked to greater default probability.
 3. Credit grades D to G exhibit significantly higher default rates.
 4. Default rates increased during the COVID-19 period.
 5. EMI-to-Income Ratio is an important indicator of financial stress.
 6. Loan purpose influences borrower risk.
 7. State-level differences in default behaviour exist.
-

8. Business Recommendations

Recommendation 1: Risk-Based Pricing

Apply higher interest rates to riskier loan categories such as small business and debt consolidation loans.

Expected Benefit

Reduced credit losses and improved profitability.

Recommendation 2: Monitor Economic Indicators

Track RBI Repo Rate changes and adjust lending policies accordingly.

Expected Benefit

Early identification of future credit stress.

Recommendation 3: Use Composite Risk Scoring

Do not rely solely on CIBIL score.

Combine:

- CIBIL Score
- Credit Utilization
- Delinquency History
- Employment Stability

Expected Benefit

More accurate borrower assessment.

Recommendation 4: Limit Exposure to High-Risk Grades

Restrict lending volume for Grades D, E, F, and G.

Expected Benefit

Reduced NPA levels.

Recommendation 5: EMI-to-Income Cap

Reject applications where the projected EMI exceeds acceptable income limits.

Expected Benefit

Lower probability of default and loss.

Conclusion

This project successfully analyzed loan portfolio data to identify major factors contributing to credit risk and loan default. Through data cleaning, exploratory data analysis, statistical testing, feature engineering, and regression modelling, several important risk patterns were discovered.

The findings suggest that borrower credit quality, loan grade, interest rate, and repayment burden significantly influence default behaviour. The recommendations provided can help FinSight Bank strengthen its credit approval process, reduce NPAs, and improve overall portfolio performance.